

Sant Gadge Baba Amravati University, Amravati
FACULTY: Science and Technology

Teaching and Learning Scheme: for the Degree of Bachelor of Science with the Major: Computer Science/Information Technology/Computer Application(Regular)/ Computer Application (Vocational)/ Data Analytics

(Three Years- Six Semesters Bachelor's Degree Programme)

(Four Years- Eight Semesters Bachelor's Degree Programme (Honors)

(Four Years- Eight Semesters Bachelor's Degree Programme (Honors with Research)

Preamble

The new curriculum of the four-year undergraduate program under NEP, for Computer Science aims to develop the core competence in computing and problem solving amongst its graduates. Informally, “Learning to learn” has been the motto of the department since its inception. The curriculum thus focuses on building theoretical foundations in Computer Science to enable its pupils to think critically when challenged with totally different and new problems. It imbibes the following **Student-Centric** features of NEP2020:

Flexibility to Exit:

In order to support early exits, the curriculum aims to develop employability skills early. This has been done so that the outcomes of the 4 yr degree is not compromised as we believe that all but a few students will go for the full 4-year degree. As programming is at the heart of computing it is proposed to have two programming courses early so that the students can develop good programming skills in the first year. At the same time students are familiarized with the hardware of computers early on.

Employability:

Industry demand in the IT sector has changed considerably in the past few years. With the humongous amount of data coming from all the domains like medical data, social networking data, astronomical data, education, etc., automating information extraction and analysis of data is the only way forward to leverage the available data for the future. The curriculum aims to equip the students with tools and techniques of Artificial Intelligence, Machine Learning and a pathway on Data Science if the student so desires. Having said this, there is no replacement for the foundational courses like programming, data structures and algorithms. With two courses on programming and three courses on data structures and algorithms together, a strong foundation will be laid down for problem solving.

Research:

With the option to obtain specialization in an area of their choice, the curriculum prepares the students to take up research projects in their final year.

Program Outcomes:

Knowledge outcomes: After completing B.Sc. Computer Science Program students will be able to:

PO1: To develop problem solving abilities using a computer.;

PO2: To prepare necessary knowledge base for research and development in Computer Science.

Skill outcomes: After completing B.Sc. Computer Science Program students will be able to:

PO3: To build the necessary skill set and analytical abilities for developing computer-based solutions.

PO4: To train students in professional skills related to Software Industry.

Generic outcomes: Students will

PO5: Augment the recent developments in the field of IT and relevant fields of Research and Development.

PO6: Enhance the scientific temper among the students so that to develop a research culture and Implementation the policies to tackle the burning issues at global and local level.

Program Specific Outcomes

PSO1: Students get knowledge and training of technical subjects so that they will be technical professional

by learning C programming, Relational Database Management, Data Structure, Software Engineering, Graphics, Java, PHP, Networking, Theoretical Computer Science, System programming, Object Oriented Software Engineering.

PSO2: Students understand the concepts of software application and projects.

PSO3: Students understand the computer subjects with demonstration of all programming and theoretical concepts with the use of ICT.

PSO4: Development of in-house applications in terms of projects

PSO5: Students will build up programming, analytical and logical thinking abilities.

PSO6: Aware them to publish their work in reputed journals

PSO7: To make them employable according to current demand of IT Industry and responsible citizen.

	1) ANSI C- Dennis Ritchie 2) Programming in C - V.Rajaraman 3) Programming with C: Venugopal K.R. TMH, Publication. 4) Programming with C: Byson Gottfried , Schaum Series Publication. Weblink to Equivalent MOOC on SWAYAM if relevant: • https://onlinecourses.swayam2.ac.in/cec19_cs06/preview • https://onlinecourses.swayam2.ac.in/nou20_cs03/preview • https://www.classcentral.com/course/swayam-computer-fundamentals-13950 • https://onlinecourses.nptel.ac.in/noc19_cs42/preview • https://onlinecourses.swayam2.ac.in/aic20_sp06/preview • https://onlinecourses.swayam2.ac.in/cec20_cs02/preview • https://www.classcentral.com/course/swayam-introduction-to-programming-in-c-2486 • https://swayamprabha.gov.in/asset/new_team/images/course_files/R12-Introduction%20to%20Programming%20in%20C%20.pdf Weblink to Equivalent Virtual Lab if relevant: • https://www.programiz.com/c-programming/online-compiler/ • https://www.onlinegdb.com/online_c_compiler • https://www.tutorialspoint.com/compile_c_online.php Any pertinent media (recorded lectures, YouTube, etc.) if relevant: • https://www.youtube.com/watch?v=eEo_aacpwCw • https://www.youtube.com/watch?v=OGM2BJ29Syg • https://www.youtube.com/playlist?list=PLWPirh4EWFpF_2T13UeEgZWZHc8nHBuXp
Model Questions:	Short Type (At least 8): 1. What do mean by Algorithm? 2. Define a flowchart. 3. What is means by program? 4. Define keyword. 5. Define Identifier. 6. Define an Array. 7. Define Structure. 8. Define Union. 9. What is a function? 10. What is String? Long Type (At least 4) 1. Describe the structure of C program. 2. Explain the looping structures in C with suitable example. 3. Describe Union and its use in C with example. 4. Illustrate Prototype of function with example. 5. Illustrate pointers with example MCQs: 1. Which of the following language is the predecessor to C Programming Language? a) A b) B c) BCPL d) C++ Ans: c 2. C programming language was developed by a) Dennis Ritchie

- b) Ken Thompson
- c) Bill Gates
- d) Peter Norton

Ans: a

3. C was developed in the year ____

- a) 1970
- b) 1972
- c) 1976
- d) 1980

Ans: b

4. C is a ____ language

- a) High Level
- b) Low Level
- c) Middle Level
- d) Machine Level

Ans: c

5. C language is available for which of the following Operating Systems?

- a) DOS
- b) Windows
- c) Unix
- d) All of these

Ans: d